

Graph-Constrained Reasoning with Bidirectional Sub-Answer Cross-Checking for Multi-Hop QA

Anonymous submission

Abstract

Some graph-enhanced reasoning variants use structural graph-validity metrics (connectivity, acyclicity, coverage) as consistency checks, but such metrics may not reflect reasoning correctness. In our structural-CS variant, the Consistency Score is weakly discriminative between correct and incorrect reasoning, and multi-path selection can degenerate toward random sampling. We propose GERS, which models reasoning as a dependency DAG executed in topological order. The core contribution is **bidirectional sub-answer cross-checking**: using the final answer and context as an anchor, each sub-question is re-derived in reverse and the forward/backward sub-answers are compared, upgrading the Consistency Score from a structural-validity metric into a content self-consistency signal. On 500 HotpotQA samples, this raises the score’s correct/wrong discrimination from -0.0035 to $+0.0847$ and AUROC from 0.498 to about 0.58 – 0.59 . The highest point-estimate configuration GERS-CV2 achieves $EM=0.302$, $F1=0.413$, improving over CoT-SC by 4 F1 points; this gain is significant under both a paired McNemar test on EM correctness ($p = 0.029$) and paired bootstrap 95% CIs that exclude zero, and GERS-CV2 also significantly outperforms a MoDeGraph-style graph-prompt baseline ($p = 0.010$). We explicitly show that self-consistency is not correctness: many high-CS answers remain wrong, and 2WikiMultiHopQA exposes a boundary on deep bridge-comparison questions where upstream entity errors propagate.

1 Introduction

Multi-hop reasoning requires models to combine evidence across multiple fragments. Chain-of-Thought (CoT) prompting (Wei et al. 2022) improves LLM performance by eliciting intermediate steps, but standard CoT organizes reasoning as **linear text** with two limitations: (1) *missing non-linear dependencies*—branching, merging, or parallel dependencies among sub-questions cannot be expressed, so a DAG-shaped path is linearized and structural information is lost; and (2) *quality-agnostic voting*—CoT-SC (Wang et al. 2023) samples multiple paths and takes the majority answer with equal-weight votes that cannot exploit path quality.

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Recent works enhance reasoning with graph structures: GoT (Besta et al. 2024) proposes graph-of-thoughts with merging/distillation but does not impose sub-question topological execution; RwG (Han et al. 2025) structures context into an entity-relation graph; MoDeGraph (Oruche et al. 2025) builds a multi-hop dependency graph as a prompt method. GERS instead studies whether graph-derived consistency scores can serve as diagnostic and weak candidate-quality signals when the graph participates in sub-question execution and checking.

We propose **GERS**, which models reasoning as a dependency DAG executed in topological order. We find that a Consistency Score computed purely from graph-theoretic properties is ineffective: LLM-generated decomposition graphs are almost always legal DAGs, so scores cluster (around 0.6 – 0.7) and multi-path selection degenerates to random sampling (discrimination only -0.0035 on 500 samples). We therefore propose the core innovation **bidirectional sub-answer cross-checking**: using the final answer and context as an anchor, each sub-question is re-derived in reverse and the forward/backward sub-answers are compared, upgrading the Consistency Score from a structural-validity metric into a content self-consistency signal. This cross-check is naturally supported by explicit DAG decomposition—which provides named sub-question units and dependency edges—and is less straightforward to apply to unstructured linear CoT. Rather than claiming uniform superiority, this work identifies and fixes a key bottleneck: structural graph validity is not equivalent to reasoning correctness. Our contributions:

1. **Mechanism (core)**: Bidirectional cross-checking raises the Consistency Score’s correct/wrong discrimination from -0.0035 to $+0.0847$ and AUROC from 0.498 to about 0.58 – 0.59 , turning consistency checking from a structural metric into a weak but useful content self-consistency signal.
2. **Empirical**: The highest point-estimate configuration GERS-CV2 achieves $EM=0.302$, $F1=0.413$ on HotpotQA (500 samples), a +4pt F1 difference over CoT-SC that is significant under both a paired McNemar test ($p = 0.029$) and paired bootstrap CIs excluding zero; it also significantly outperforms a MoDeGraph-style baseline ($p = 0.010$).
3. **Diagnostic**: A boundary analysis on 2WikiMultiHopQA

shows the method suits comparison-style decomposition but fails on deep bridge-comparison questions where upstream errors propagate. The tested structural-score graph-level self-consistency yields no extra gain.

2 Related Work

Chain-of-Thought reasoning. Wei et al. (2022) proposed CoT prompting. Wang et al. (2023) proposed Self-Consistency (CoT-SC), whose equal-weight voting does not exploit path quality. Yao et al. (2023) proposed Tree of Thoughts (ToT), with no explicit dependency modeling among sub-questions. These methods are based on linear or tree structures.

Graph-enhanced reasoning. GoT (Besta et al. 2024) models reasoning as a graph for merging/distillation but focuses on prompt engineering. RwG (Han et al. 2025) uses a context graph as a one-time input. MoDeGraph (Oruche et al. 2025) builds a multi-hop dependency graph but remains a prompt method. GERS couples the reasoning graph with execution and studies graph-derived consistency as a diagnostic and weak ranking signal.

Consistency checking and self-evaluation. Process Reward Models (Lightman et al. 2024) score intermediate steps but require large amounts of labeled data. Self-Refine (Madaan et al. 2023) iteratively refines answers via self-feedback but operates on linear text without a separately checkable sub-question structure. GERS uses graph-theoretic algorithms for structural checking and designs bidirectional cross-checking so that consistency scores can be analyzed and used in controlled variants. A qualitative positioning table against these families is given in the supplement.

3 Method

Overall Framework

GERS has three modules: reasoning-state graph representation, graph-constrained chain generation, and consistency checking. The flow (Figure 1): input $Q, C \rightarrow$ decompose into sub-questions $\{q_i\}$ with dependencies $\rightarrow$ build DAG $G = (V, E) \rightarrow$ topological sort $\tau \rightarrow$ execute step-by-step to get sub-answers $a_i \rightarrow$ aggregate into final answer A (answer-type alignment) $\rightarrow$ compute the cross-validated consistency score $S = 0.3 \cdot S_{struct} + 0.7 \cdot S_{crossval} \rightarrow$ output A, G, S . We also evaluate a GERS-SC diagnostic baseline that samples K DAGs and selects by the original structural CS.

Reasoning-State Graph Representation

Nodes: FactNode (facts/original question), StepNode (sub-question and answer), ConclusionNode (final conclusion). Edges: DeriveEdge ($u \rightarrow v$), SupportEdge, ConflictEdge. The LLM decomposes Q into an ordered sub-question list with dependencies, and nodes/edges are created accordingly.

Graph-Constrained Chain Generation

Topological sort gives execution order $\tau = (v_1, \dots, v_n)$; cycles are broken first, so each sub-question is answered only after its predecessors. Sub-questions are answered in topological order with each answer passed as context to dependent

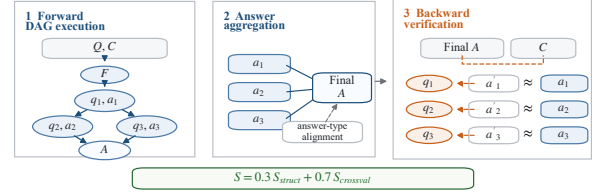


Figure 1: GERS reasoning framework: decompose into a dependency DAG, execute sub-questions in topological order, aggregate a final answer, and perform bidirectional sub-answer cross-checking anchored by the final answer and context.

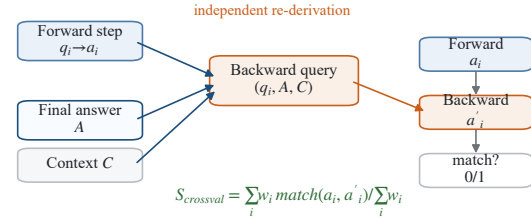


Figure 2: Bidirectional sub-answer cross-checking: each sub-question is re-derived in reverse from the answer+context; forward/backward agreement yields a content self-consistency score that replaces the non-discriminative structural score.

sub-questions (dependency propagation), then the chain is aggregated into A . **Answer-type alignment** constrains A to match the entity type requested by the original question (e.g., “which film” $\rightarrow$ a film name, not an intermediate director), which is critical for bridging-comparison questions.

Consistency Checking and Bidirectional Cross-Validation

Structural Consistency Score (baseline).

$$S_{struct} = w_1 \cdot C_{conn} + w_2 \cdot C_{cycle} + w_3 \cdot C_{cov} \quad (1)$$

with $w_1 = w_3 = 0.35, w_2 = 0.30$ for connectivity, acyclicity, coverage, plus a chain-length penalty. LLM-generated graphs are almost always legal connected acyclic DAGs, so S_{struct} clusters (mean 0.662, std 0.098, discrimination -0.0035) and cannot distinguish reasoning quality.

Sub-answer bidirectional cross-validation (core). Given forward sub-answers $\{a_i\}$ and final answer A , each sub-question is re-answered in reverse using “ A + context C ” as the anchor, yielding backward sub-answers $\{a'_i\}$:

$$S_{crossval} = \frac{\sum_{i=1}^n w_i \cdot 1[\text{match}(a_i, a'_i)]}{\sum_{i=1}^n w_i} \quad (2)$$

where $w_i = 0.5 + 0.5 \cdot (i + 1)/n$ gives downstream sub-questions higher weight, as their errors propagate further. The match function is a three-level heuristic: (i) exact match after normalization (lowercasing, punctuation removal, yes/no and

numeric unification); (ii) substring containment or numeric equality; (iii) otherwise 0. An optional LLM judge (disabled by default for cost) handles residual semantic cases. The reverse prompt requires “re-derive from the context, do not copy the final answer” to mitigate confirmation bias. This cross-check is naturally supported by explicit DAG decomposition and is more interpretable than applying it to unstructured CoT.

Confirmation-bias control. A context-only backward variant that removes the answer anchor obtains EM 0.300 / F1 0.414 with lower CS 0.646, close to the default EM 0.302 / F1 0.413 / CS 0.777, suggesting the end-task gain is not driven by simply feeding the final answer back.

New Consistency Score. The layers are combined with fixed weights (set heuristically, held constant across all experiments, no test-set tuning):

$$S = 0.3 \cdot S_{struct} + 0.7 \cdot S_{crossval} \quad (3)$$

Confidence-Weighted Aggregation

Each sub-answer confidence is estimated by a lightweight heuristic (no extra LLM call): non-emptiness, absence of uncertainty signals, entity grounding in context, and pure-number check. Sub-answers below $\theta_{conf} = 0.5$ are tagged [LOW CONFIDENCE] in the aggregation prompt, reducing single-step error propagation.

Graph-Level Self-Consistency (GERS-SC)

GERS-SC generates K DAGs (decomposition temperatures $T \in \{0.3, 0.5, 0.7\}$) and selects the highest-scoring one. The main-table variant uses the original structural CS for selection; its failure motivates bidirectional cross-validation. Full multi-path selection with the new cross-validated score is left for future work.

4 Experiments

Setup

HotpotQA (Yang et al. 2018) (500 samples: 404 bridge + 96 comparison) is the main benchmark. 2WikiMultiHopQA (Ho et al. 2020) is used as a boundary analysis (100-sample typed diagnostic split plus a 300-sample larger check, reported in the supplement). Baselines: Zero-Shot, Standard CoT, CoT-SC ($N = 3$), CoT-SC+GERS rerank (ablation), and a MoDeGraph-style graph-prompt baseline—an unofficial implementation from the public method description, not an official reproduction. All methods use a unified concise answer extraction/normalization pipeline. Our methods: GERS+adaptive (DAG execution + adaptive routing); GERS-SC (+graph-level self-consistency $K = 3$); GERS-CV (+bidirectional cross-validation); GERS-CV2 (+confidence weighting, highest point estimate). Model: Qwen3-8B (DashScope API), 4-worker parallel, zero failures. Metrics: EM, F1, Consistency Score. Main comparisons report paired bootstrap 95% CIs (10000 resamples of per-sample differences) and McNemar tests on paired EM correctness. To reduce evaluation-pipeline distortions we unify three treatments: fix an EM bidirectional-substring bug, use the same extraction/normalization code for all methods, and add answer-type alignment in GERS aggregation.

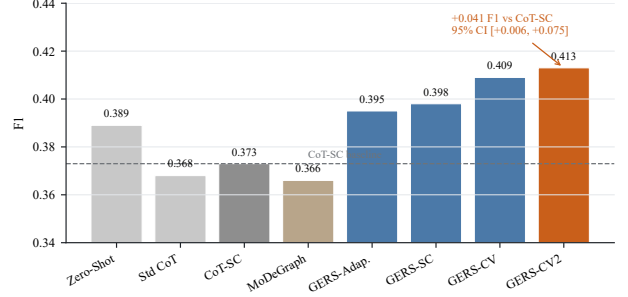


Figure 3: Main F1 results on HotpotQA (n=500). GERS-CV2 has the highest point-estimate F1; the annotation reports the paired bootstrap interval for the F1 difference against CoT-SC.

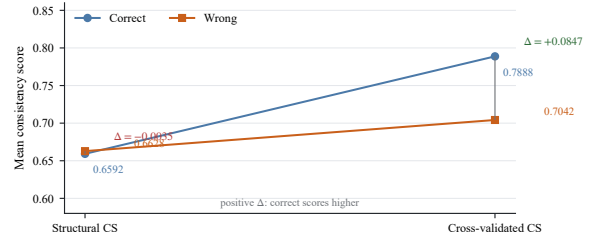


Figure 4: Consistency Score discrimination (HotpotQA, n=500). The old structural CS clusters around 0.66 for both correct and wrong answers (discrimination -0.0035); the new CS with bidirectional cross-checking separates them weakly ($+0.0847$), on the GERS-CV variant.

Main Results

GERS-CV2 has the highest point-estimate EM=0.302, F1=0.413, with a +4pt F1 difference over CoT-SC. It also significantly outperforms the MoDeGraph-style baseline (F1 0.366; McNemar $p = 0.010$, paired F1 CI $[+0.012, +0.083]$). Bidirectional cross-checking ($0.284 \rightarrow 0.298$) is associated with a +1.4pt EM change. The CoT-SC+GERS rerank (0.372) is on par with CoT-SC (0.373), so applying graph scores post hoc to already-generated linear CoT answers is insufficient. Zero-Shot (0.276) is already close to GERS on HotpotQA—its context directly contains answer evidence, compressing the advantage of structured methods (analyzed in Section 4.4).

GERS-CV2 yields a significant paired McNemar test against CoT-SC ($p = 0.029$), and the paired bootstrap CIs for both EM ($[+0.006, +0.074]$) and F1 ($[+0.006, +0.075]$) exclude zero: the end-task improvement is small in magnitude but statistically significant under paired evaluation. The same holds against Standard CoT ($p = 0.040$) and the MoDeGraph-style baseline ($p = 0.010$). By contrast, GERS-SC without cross-checking ($p = 0.275$; CI crossing zero) is not significant, corroborating that the gain stems from bidirectional cross-checking.

Table 1: Main results (HotpotQA, n=500)

Method	EM	F1	CS
Zero-Shot	0.276	0.389	—
Std CoT	0.264	0.368	—
CoT-SC (N=3)	0.262	0.373	—
CoT-SC+GERS rerank	0.264	0.372	—
MoDeGraph-style prompt	0.252	0.366	—
GERS-Adap.	0.284	0.395	0.996
GERS-SC (K=3)	0.282	0.398	0.662
GERS-CV (+cross-check)	0.298	0.409	0.782
GERS-CV2 (+confidence)	0.302	0.413	0.777

Table 2: Significance tests (HotpotQA, n=500). EM/F1 differences report paired bootstrap 95% CIs (10000 resamples of per-sample differences); McNemar p on paired EM correctness. CIs excluding zero indicate a significant paired difference.

Comparison	EM diff	EM 95% CI	F1 diff	F1 95% CI	McNemar p
GERS-CV2 vs CoT-SC	+0.040	[+0.006, +0.074]	+0.041	[+0.006, +0.075]	0.029
GERS-CV2 vs Std CoT	+0.038	[+0.004, +0.072]	+0.045	[+0.010, +0.081]	0.040
GERS-CV2 vs MoDeGraph	+0.050	[+0.014, +0.086]	+0.047	[+0.012, +0.083]	0.010
GERS-SC vs CoT-SC	+0.020	[−0.012, +0.052]	+0.025	[−0.009, +0.060]	0.275

Table 3: CS correct/wrong discrimination (HotpotQA, n=500). AUROC uses EM correctness as the binary label.

CS scheme	Correct	Wrong	Disc.	AUC
Old CS (structural, GERS-SC)	0.6592	0.6628	−0.0035	0.4
New CS (+cross-val, GERS-CV)	0.7888	0.7042	+0.0847	0.5
New CS (GERS-CV2)	—	—	—	0.5

Table 4: 2WikiMultiHopQA per-type F1 (n=100)

Method	comp.	bridge_c	compos.	infer.
Standard CoT	0.815	0.905	0.284	0.361
CoT-SC	0.720	0.864	0.268	0.343
GERS-CV2	0.775	0.524	0.297	0.288

Repairing Consistency Score Discrimination (Core Evidence)

The old CS provides little discriminative signal (discrimination −0.0035, reverse; clustered). Bidirectional cross-checking repairs the direction and magnitude (+0.0847; spread). The AUROC rises from 0.498 to 0.58–0.59, so cross-checking makes CS more informative than structural validity. However, 156 of the 250 samples with CS=1.0 are still EM-wrong; internal agreement is not factual correctness. The signal is better read as a weak ranking/diagnostic signal than a correctness verifier (a CS-bucket breakdown is in the supplement).

Per-Type Analysis and Boundary

On *comparison*, GERS-CV2 (0.775) is close to Standard CoT; the DAG naturally decomposes the two comparison

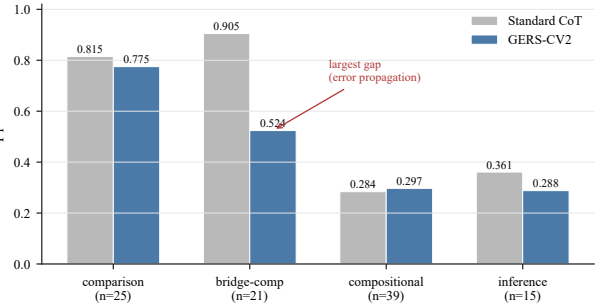


Figure 5: 2WikiMultiHopQA per-type F1 (n=100, diagnostic). GERS-CV2 is close to CoT on comparison but lags on bridge_comparison due to upstream entity error propagation.

branches. On *bridge_comparison*, GERS’s clear weakness (0.524 vs 0.905): this “bridge → compare → return to original entity” composite suffers from sub-question error propagation. Cross-validation raises it from 0.476 (without CV) to 0.524—partial mitigation, not a cure. The tested GERS-SC (0.282) ≈ GERS+adaptive (0.284), $p = 1.000$: structural-score multi-path selection is ineffective, not a result about all cross-validated multi-path schemes. The advantage of GERS is concentrated on comparison/branch-merge samples; on bridge questions all methods are close because Qwen3-8B extracts answers directly from context. A 300-sample 2Wiki check and a Qwen-Plus generalization run (where the structured advantage disappears) are reported in the supplement.

Ablation

Cross-checking is the main observed gain source (0.284→0.298, +1.4pt EM; CS discrimination −0.0035 → +0.0847). Structural-score graph-level self-consistency

Table 5: Ablation (HotpotQA, n=500)

Config	EM	F1	Note
GERS+adaptive	0.284	0.395	baseline: DAG + adaptive
GERS-SC (+K=3)	0.282	0.398	multi-path, old CS
GERS-CV (+cross-val)	0.298	0.409	new CS discriminative
GERS-CV2 (+conf.)	0.302	0.413	highest point est.
w/o graph exec.	0.264	0.372	no DAG execution

Table 6: Real case study: evidence cue and forward/backward agreement (abbreviated).

Case	Evidence cue	Forward a_i	Backward a'_i	MatchNew CS
A	Derrickson/Wood	American/American	American/American	1.0
B	context lacks 1922, 2014, claimed years ...	"not in con- text"	×	0.0

yields no extra gain ($p = 1.000$). Confidence weighting contributes marginally (+0.4pt, not significant). Graph execution is the foundation: removing it degrades to 0.264. A cost analysis (GERS-CV2 ~6–8 LLM calls, 6.6s) is in the supplement.

Case Study

Case A (self-consistent, correct). “Were Scott Derrickson and Ed Wood of the same nationality?” Forward decomposes into two nationality sub-questions, both “American” → “yes”. Reverse verification, anchored on “yes”+context, independently re-derives “American”/“American”, matching forward; crossval=1.0.

Case B (inconsistent, hallucination caught). “Which director had the longest career, Alain Resnais or Scott Sidney?” (gold: Scott Sidney). Forward hallucinates birth/death years (1922, 2014, 1902, 1975) absent from the context and concludes “Alain Resnais”. Reverse verification, re-deriving from context, returns “not in context” for all 7 sub-questions; crossval drops to 0.0. The old structural CS stays 0.7 (legal DAG) and detects nothing; the new CS reflects the content inconsistency.

5 Discussion and Limitations

Core value. The main contribution is raising CS discrimination from -0.0035 to $+0.0847$ (AUROC $0.498 \rightarrow 0.58\text{--}0.59$), upgrading consistency checking from “is the graph legal” to “is the reasoning content self-consistent”. The EM gain is modest and the score is still a weak correctness predictor; its value is diagnostic and ranking-oriented.

Self-consistency is not correctness. Raising the cross-validation score can make a chain more internally self-consistent without making it more correct: CS=1.0 still contains many EM-wrong answers, and repair variants raise CS while reducing EM/F1 (supplement). Bidirectional cross-checking is a content-consistency signal, not a complete correctness verifier.

Honest positioning. GERS-CV2 is significantly better than CoT-SC, Standard CoT, and the MoDeGraph-style baseline on HotpotQA EM under paired evaluation (McNemar $p \leq 0.040$; CIs excluding zero), but the gain is small in magnitude (+4pt F1) because HotpotQA’s context directly contains answer evidence.

Limitations. On bridge_comparison, GERS’s error propagation lags behind linear CoT—an inherent limitation of graph decomposition on deep multi-hop. The MoDeGraph-style baseline is an unofficial implementation; the comparison covers one prompt-only graph baseline, not all graph-reasoning systems.

6 Conclusion

We propose GERS, whose core innovation is **bidirectional sub-answer cross-checking**: re-deriving sub-questions in reverse from the answer+context and comparing forward/backward consistency, upgrading the Consistency Score from a pure graph-theoretic metric (discrimination -0.0035 , AUROC 0.498) to a more informative content self-consistency signal ($+0.0847$, AUROC 0.58–0.59), naturally supported by explicit DAG decomposition. GERS-CV2 achieves EM=0.302, F1=0.413 on 500 HotpotQA samples, a +4pt F1 difference over CoT-SC that is significant under both a paired McNemar test ($p = 0.029$) and paired bootstrap CIs excluding zero, and a significant improvement over a MoDeGraph-style baseline ($p = 0.010$). We also provide a boundary analysis showing the method suits comparison-style decomposition but fails on deep bridge-comparison questions. GERS should be read as a step toward interpretable graph-based reasoning diagnostics, not a solved correctness verifier.

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